Adversarial Attacks

Robust Deep Learning (Seminar)

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Which one is the gibbon?

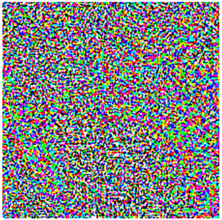
Which one is the gibbon?

panda

gibbon

A fragile foundation

$+ .007 \times$



$=$

$\boldsymbol{x}$

"panda"
57.7% confidence

$\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$

"nematode"
8.2% confidence

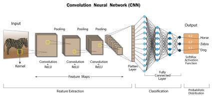
$\boldsymbol{x} + \epsilon\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$

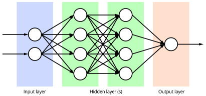
"gibbon"
99.3 % confidence

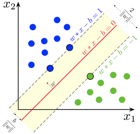
Goodfellow et al., 2014

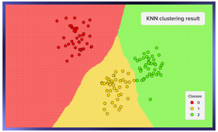
Transferability

- An adversarial example crafted against one model often fools other models as well









Papernot et al., 2016

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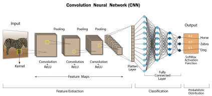

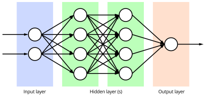

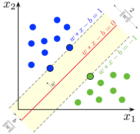

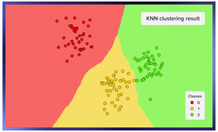

Papernot et al., 2016

Transferability

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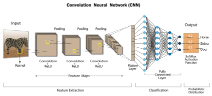
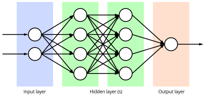
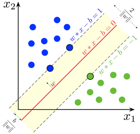
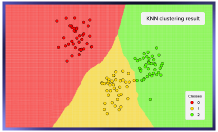
- Survives a change of architecture as well as of training data, initialisation, hyperparameters
- Vulnerability apparently *inherent to problem itself*, not to one model or one architecture









Papernot et al., 2016

Why this matters

- **Fundamental blind spot**: model and human do not share one understanding of input

Images: medicalimaging.na; evsmarts.com; thetravelimages.com

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- Raises questions about **reliability** of systems

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Why this matters

- **Fundamental blind spot**: model and human do not share one understanding of input
- Raises questions about **reliability** of systems
- More pressing as systems are entering deeper and deeper into **everyday life** and critical infrastructure

Images: medicalimaging.na; evsmarts.com; thetravelimages.com

Outline

Outline

1. What is an adversarial example?
2. Taxonomy
3. Evasion attacks
4. Poisoning attacks
5. Discussion

What is an Adversarial Example?

Definition (Adversarial Example)

An *adversarial example* is an **input to a machine learning model** that has been intentionally perturbed, typically **in a subtle and often imperceptible manner**, such that the model **produces an incorrect prediction** while the semantic content of the input remains unchanged.

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- **Machine learning models**: No quirk of deep networks, but a general property of machine learners
- **Subtle and imperceptible perturbations**: mathematically small under a chosen norm / visually imperceptible to humans
- **Produces an incorrect predition**: The machine learner is fooled where the human would not have been

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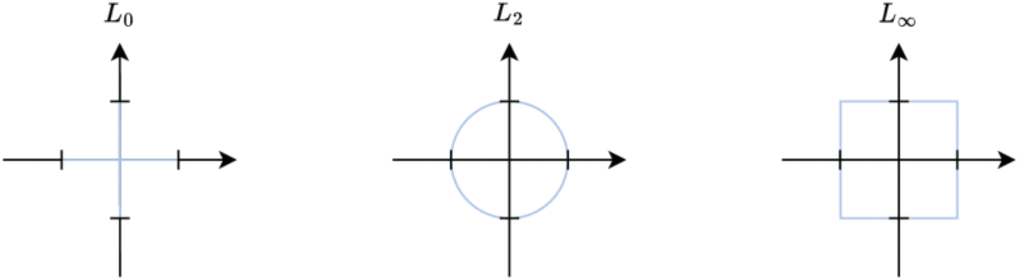
Four orthogonal axes

- **Attacker knowledge**: White-box vs. Black-box
- **Attack stage**: Evasion (at testing time) vs. Poisoning (at training time)
- **Attack goal**: Targeted (specific target label) vs. Untargeted (just any wrong label)
- **Perturbation constraint**: which Norm, which Budget

Measuring "small": the ℓ_p norms



$$||x||_0 = (|x_1|^0 + \ldots + |x_n|^0) \qquad ||x||_2 = (|x_1|^2 + \ldots + |x_n|^2)^{\frac{1}{2}} \qquad ||x||_\infty = \max_i |x_i|$$

Norm defines the **perturbation budget** ϵ.

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L-BFGS

Szegedy et al., 2013

White-box · Evasion · Targeted · ℓ_2

$$\min_{\delta} \|\delta\|_2 \qquad \text{subject to} \qquad f(x+\delta) = t$$

- f has no gradient — it returns a discrete class label, $f(x+\delta) = t$ is a hard, non-differentiable condition.
- Derivative-based optimisers such as L-BFGS need a gradient to move at all.

$\rightsquigarrow$ Reformulate.

L-BFGS
Szegedy et al., 2013

White-box · Evasion · Targeted · ℓ_2

$$\min_{\delta} \; c \, \|\delta\|_2 + L_f(x + \delta, \; t)$$

Significance

- First method to **reliably generate** adversarial examples.
- Computationally expensive — far too slow to be practical.

$\leadsto$ It *proved a point* rather than providing a usable attack.

The arc of the field

2013 **L-BFGS** — expensive, but proves concept

Fast Gradient Sign Method

Goodfellow et al., 2014

White-box · Evasion · Untargeted · ℓ_∞

- Previously: Adversarial examples assumed to be caused by 'complex' behaviour
- Goodfellow et al.: Linear models have adv. ex. as well

$\leadsto$ Models may be too linear

Do Neural Networks behave linearly?

Goodfellow et al., 2014

...

Fast Gradient Sign Method

Goodfellow et al., 2014

White-box · Evasion · Untargeted · ℓ_∞

- Linearise the loss function:

$$\arg\max_{||\delta||_\infty \le \epsilon} L_f(x+\delta, y) = \arg\max_{||\delta||_\infty \le \epsilon} L_f(x, y) + \nabla_x L_f(x, y)^T \delta$$

- Maximised by $\delta = \epsilon \cdot sign(\nabla_x L_f(x, y))$

$\rightsquigarrow$ single step in gradient direction

Fast Gradient Sign Method - Example

Goodfellow et al., 2014

White-box · Evasion · Untargeted · ℓ_∞

placeholder: step graph here

[Results/Significance]

Goodfellow et al., 2014

White-box · Evasion · Untargeted · ℓ_∞

- Maximised by $\delta = \epsilon \cdot sign(\nabla_x L_f(x, y))$

$\rightsquigarrow$ single step in gradient direction

PGD
Madry et al., 2017

White-box · Evasion · ??? · ???

General formulation of adversarial training [...! incl. defenders part?]. Adversary solves

$$\max_{\delta \in S} L_f(x + \delta, y)$$

- FGSM: Use $S = \{\delta \mid \|\delta\|_\infty \leq \epsilon\}$
- Use projected gradient descent to optimise

$\leadsto$ Models may be too linear

PGD - a general first-order attack?
Madry et al., 2017

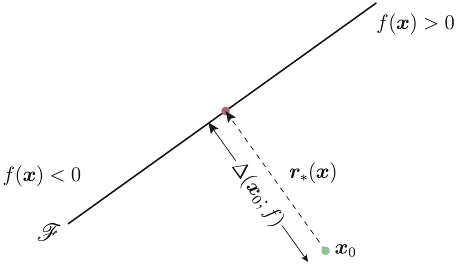
White-box · Evasion · ??? · ???

White-box · Evasion · Untargeted · ℓ_2

DeepFool

Moosavi-Dezfooli et al., 2016 — the affine, binary case

- For an affine $f(x) = w^\top x + b$, the decision boundary $\mathcal{F}$ is a *hyperplane*
- Minimal perturbation is *orthogonal projection* of x onto $\mathcal{F}$ $\rightarrow$ closed form

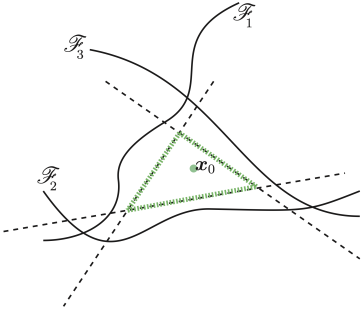


DeepFool

Moosavi-Dezfooli et al., 2016 — the general, multiclass case

White-box · Evasion · Untargeted · ℓ_2

- **Non-linear:** iterative $\rightarrow$ linearise the score function at current point (first-order Taylor), project onto that hyperplane, repeat until label flips.
- **Multiclass:** Linearise *every* class boundary, project onto the **closest**.



Significance

- Gave problem a **geometric reading**: robustness as distance to the decision boundary.
- First to **extend to any ℓ_p norm**, $p \in [1, \infty)$.

The arc of the field

2013 **L-BFGS** — expensive, but proves concept

2016 **DeepFool** — minimal perturbation via iterative linear approximation

Carlini & Wagner

Carlini & Wagner, 2017

White-box · Evasion · Targeted · $\ell_0/\ell_2/\ell_\infty$

$$\underbrace{\min_{\delta}\ c\,\|\delta\|_2 + L_f(x+\delta,\ t)}_{\text{L-BFGS}} \quad\longrightarrow\quad \underbrace{\min_{\delta}\ \|\delta\|_p + c\cdot\ell(x+\delta)}_{\text{C\&W}}$$

- $\ell(x') = \max\big(\max_{i\neq t} z_i(x') - z_t(x'),\ 0\big).$
- Built so that $\ell(x') \leq 0 \iff$ input is classified as $t \rightarrow$ faithfully encodes the attack goal.
- c decided through **binary search**

Carlini & Wagner

Carlini & Wagner, 2017

White-box · Evasion · Targeted · $\ell_0/\ell_2/\ell_\infty$

$$x' = \tfrac{1}{2}\big(\tanh(w) + 1\big), \qquad w \in \mathbb{R}^m$$

- mapping any $w \in \mathbb{R}^m$ into $(0,1)^m$ $\rightarrow$ every w decodes to a **valid image** $\rightarrow$ dissolving box-constraint
- Optimise w **unconstrained** with Adam optimiser

Significance

- Strongest attack of its time across $\ell_0/\ell_2/\ell_\infty$.
- Broke *defensive distillation*, exposing it as *gradient masking*

The arc of the field

2013 **L-BFGS** — expensive, but proves concept

2016 **DeepFool** — minimal perturbation via iterative linear approximation

2017 **C&W** — reformulate optimisation

Transfer attacks

- Black-box attack pattern
- Adversary creates and trains a substitute model for the target
- Adversarial examples for substitute model are found (white-box!)
- These can be used to attack the target

Jacobian-Based Dataset Augmentation

Papernot et al., 2017

Black-box · Evasion · ??? · ???

- Assumption: No knowledge of model parameters, dataset
  Access to somewhat representative inputs is granted (e.g. natural images)
- How do we construct a suitable substitute?
- Construct an artificial dataset!

Jacobian-Based Dataset Augmentation

Papernot et al., 2017

- Assumption: No knowledge of model parameters, dataset
  Access to somewhat representative inputs is granted (e.g. natural images)
- How do we construct a suitable substitute? (similar decision boundaries)
- Construct an artificial dataset!

Jacobian-Based Dataset Augmentation

Papernot et al., 2017

- Use target model as an oracle to label samples
- Strategy for sample selection is necessary
- Idea: iteratively add meaningful samples to dataset
- Use Jacobian for a heuristic

Heuristic for meaningful directions

Papernot et al., 2017

$$x = x + \alpha \cdot sign(J_f(x)[o(x)])$$

- $J_f(x)$: total derivative
- $J_f(x)[o(x)]$: total derivative, column of the correct label

$\leadsto$ We move into the class region (according to the substitute)!

Jacobian-Based Dataset Augmentation

Papernot et al., 2017

- 1) collect samples somewhat related to the input domain
- 2) choose architecture for substitute
- 3) target model labels current samples
- 4) train substitute
- 5) expand dataset: $S_{N+1} := \{x = x + \alpha \cdot sign(J_f(x)[o(x)]) \mid x \in S_N\} \cup S_N$
- repeat (3) onwards

White-box attacks to improve transferability

- Adversarial examples can differ in transferability!
- MI-FGSM: adds momentum to FGSM
- Ensemble attacks

Outline

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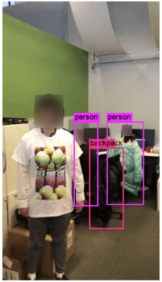
Physical adversarial attacks



Sharif et al., 2016 · Xu et al., 2020

Eykholt et al., 2018

Athalye et al., 2018

Real-life adversarial attacks

- Real confirmed malicious attacks are surprisingly rare

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- Two candidate explanations:
    1. They slip through unnoticed, logged as ordinary misclassifications
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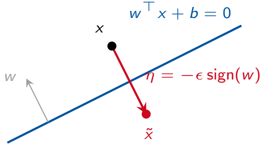
Real-life adversarial attacks

- Real confirmed malicious attacks are surprisingly rare
- Two candidate explanations:
  1. They slip through unnoticed, logged as ordinary misclassifications
  2. They are simply of little use to actual attackers
- Case for the second:
  - Needs ML *and* security expertise
  - DDoS, character obfuscation, packing a binary etc. $\rightarrow$ cheap, fast, effective

Explaining adversarial examples

High-dimensional linearity
Goodfellow et al., 2014

- Affine case first: perturbation shifts activation by $w^\top \eta$



$$w^\top \tilde{x} = w^\top (x + \eta) = w^\top x + w^\top \eta$$

$$w^\top \eta = \epsilon \sum_{i=1}^{n} |w_i| = \epsilon\, m\, n$$

n dimensions, mean weight magnitude m

High-dimensional linearity

Goodfellow et al., 2014

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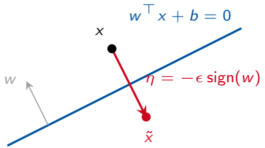
- Best strategy: η aligned with the weights



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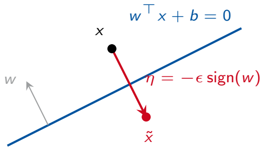

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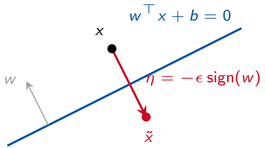
$\leadsto$ Many imperceptible changes accumulate into one large shift



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From linear models to deep networks

Goodfellow et al., 2014

- Most DNN architectures behave *locally linearly* for ease of optimisation

ReLU

LSTM

sigmoid near zero

maxout

linear-softmax

From linear models to deep networks

Goodfellow et al., 2014

- Most DNN architectures behave *locally linearly* for ease of optimisation

$\rightsquigarrow$ Within ϵ-ball, the linear argument carries over

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LSTM

maxout

sigmoid near zero

linear-softmax

Predictive and robust features

Ilyas et al., 2019

Predictive

Correlated with the label in expectation.

Predictive and robust features

Ilyas et al., 2019

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Robust

Stays correlated *under* adversarial perturbation.

Predictive and robust features
Ilyas et al., 2019

Predictive
Correlated with the label in expectation.

Robust
Stays correlated *under* adversarial perturbation.

$\leadsto$ **Predictive but non-robust:** Standard training happily relies on such features.

Non-robust features are real signal
Ilyas et al., 2019

Why that works

Ilyas et al., 2019

	Robust features point at	**Non-robust features point at**	**Label**	**Image**
Original clean training image (in $\mathcal{D}$)	dog	dog	dog	
Perturbed training image (in $\widehat{\mathcal{D}}_{NR}$)	dog *(unaffected by perturbation)*	cat *(through adversarial perturbation)*	cat	
Original clean test image (in $\mathcal{D}$)	cat	cat	**cat** (predicted)	

Adversarial attacks as a subject of study

- Not really significant security threat

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  - Not: What can an attacker do?
  - But: What do they reveal about the targetted system?
- Before
  - Accuracy as near-single metric
  - Robustness assumed implicitly
- Afterwards:
  - High accuracy $\neq$ understanding the task as a human would
  - A more intricate account of how learners operate
  - Robustness as a distinct property of interest

Thank you for your attention!

Questions?

References I